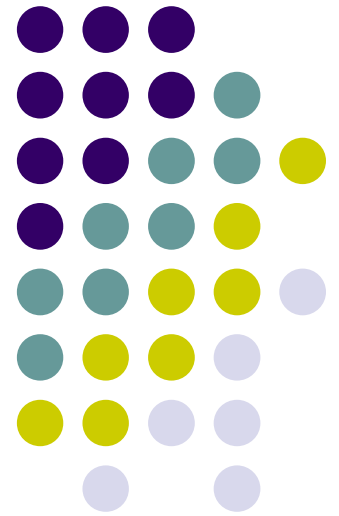
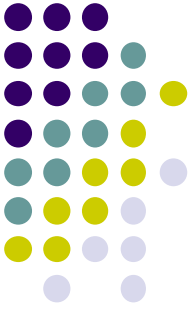


# WIA1005 Network Technology Foundation

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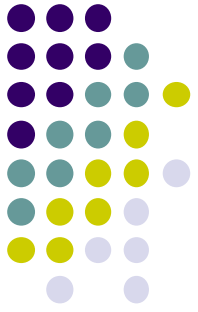
## Chapter 11 Routing





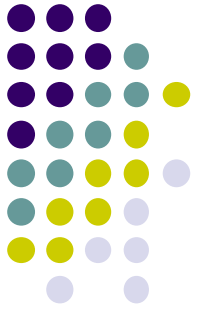
# Contents

- Introduction
- Packet Forwarding
- Routing Table
- Static Route
- Dynamic Route



# Introduction

- A router connects multiple networks, which means that it has multiple interfaces that each belong to a different IP network.
- The router is responsible for the routing of traffic between networks. When a packet arrives on a router interface, the router uses its routing table to determine how to reach the destination network.
- The primary functions of a router are to:
  - Determine the best path to forward packets based on the information in its routing table
  - Forward packets toward their destination



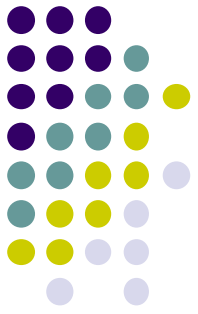
# Introduction

- The best path in the routing table is also known as the longest match. The longest match is the route in the routing table that has the greatest number of far-left matching bits with the destination IP address of the packet.

| Destination IPv4 Address |                      | Address in Binary                   |
|--------------------------|----------------------|-------------------------------------|
| 172.16.0.10              |                      | 10101100.00010000.00000000.00001010 |
| Route Entry              | Prefix/Prefix Length | Address in Binary                   |
| 1                        | 172.16.0.0/12        | 10101100.00010000.00000000.00001010 |
| 2                        | 172.16.0.0/18        | 10101100.00010000.00000000.00001010 |
| 3                        | 172.16.0.0/26        | 10101100.00010000.00000000.00001010 |

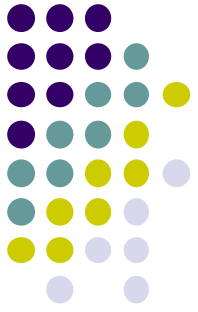
For the destination IPv6 packet with the address **2001:db8:c000::99**

| Route Entry | Prefix/Prefix Length    | Does it match?                   |
|-------------|-------------------------|----------------------------------|
| 1           | 2001:db8:c000::/40      | Match of 40 bits                 |
| 2           | 2001:db8:c000::/48      | Match of 48 bits (longest match) |
| 3           | 2001:db8:c000:5555::/64 | Does not match 64 bits           |



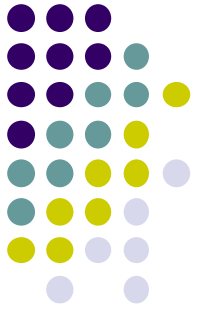
# Introduction

- The routing table of a router stores information about:
  - **Directly connected routes** - These routes come from the **active router interfaces** directly connected to the router.
  - **Remote routes** - These are remote networks connected to other routers. Routes to these networks can either be statically configured or dynamically configured using dynamic routing protocols.
  - **Default route** - Specifies a next-hop router to use when the routing table does not contain a specific route that matches the destination IP address. The default route can be entered manually or learned automatically from a dynamic routing protocol.



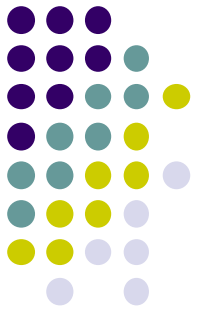
# Introduction

- Determining the best path involves the evaluation of multiple paths to the same destination network and selecting the **optimum or shortest path** to reach that network.
- The best path is selected by a routing protocol based on the value or metric it uses to determine the distance to reach a network. A metric is the quantitative value used to measure the distance to a given network. **The best path to a network is the path with the lowest metric.**
- When a router has two or more paths to a destination with **equal cost metrics**, then the router forwards the packets using both paths equally. This is called equal cost load balancing.



# Introduction

- Different routing protocols use different metric
  - **Routing Information Protocol (RIP)** - Hop count
  - **Open Shortest Path First (OSPF)** - Cost (cumulative bandwidth from source to destination)
  - **Enhanced Interior Gateway Routing Protocol (EIGRP)** - Bandwidth, delay, load, reliability



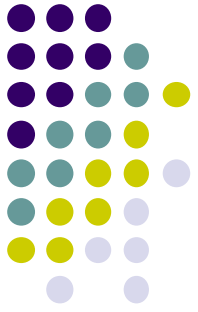
# Introduction

- Cisco IOS uses the administrative distance (AD) to determine the route to install into the IP routing table.
- The AD represents the "trustworthiness" of the route; **the lower the AD, the more trustworthy the route source.**

Default Administrative Distances

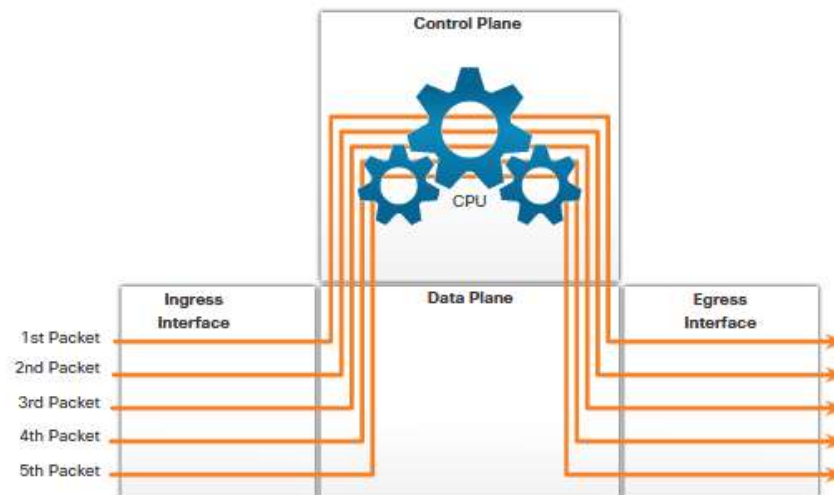
| Route Source        | Administrative Distance |
|---------------------|-------------------------|
| Connected           | 0                       |
| Static              | 1                       |
| EIGRP summary route | 5                       |
| External BGP        | 20                      |
| Internal EIGRP      | 90                      |
| IGRP                | 100                     |
| OSPF                | 110                     |
| IS-IS               | 115                     |
| RIP                 | 120                     |
| External EIGRP      | 170                     |
| Internal BGP        | 200                     |

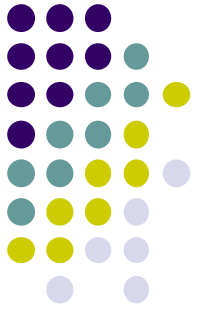




# Packet Forwarding

- Routers support three packet-forwarding mechanisms:
- **Process switching**
  - When a packet arrives on an interface, it is forwarded to the control plane where the CPU matches the destination address with an entry in its routing table, and then determines the exit interface and forwards the packet. Process every packet, very slow.

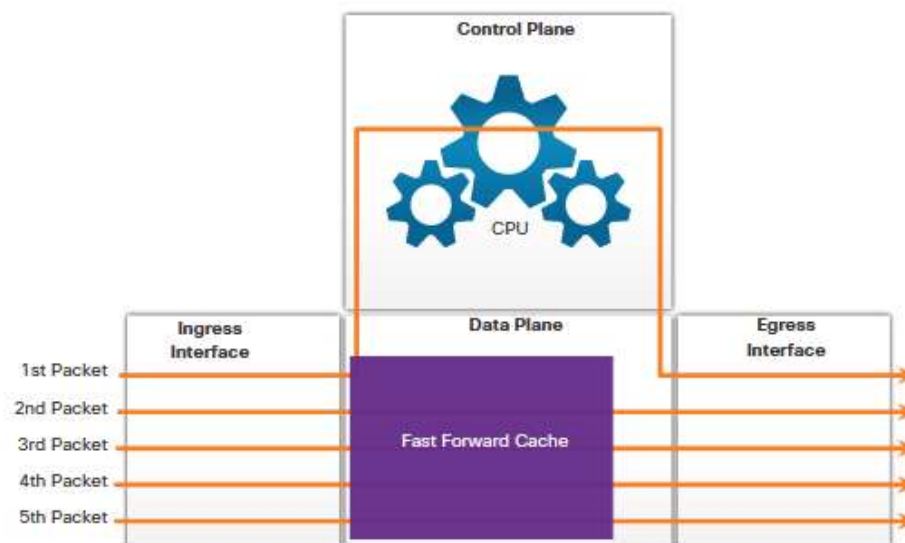


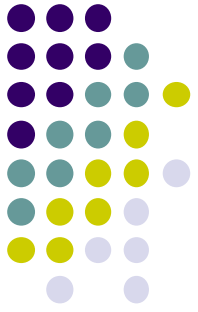


# Packet Forwarding

- **Fast switching**

- Uses a fast-switching cache to store next-hop information. If another packet going to the same destination arrives on an interface, the next-hop information in the cache is re-used without CPU intervention.

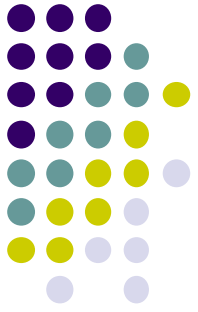




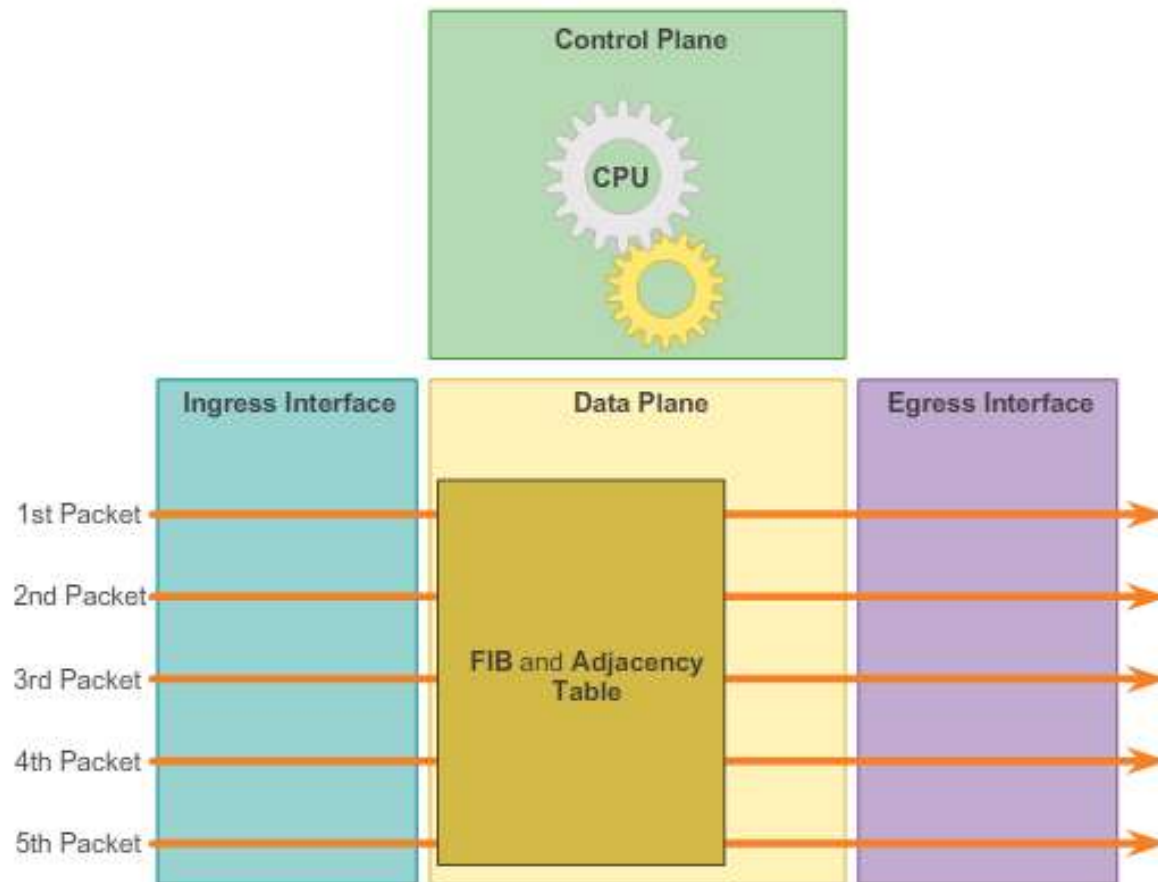
# Packet Forwarding

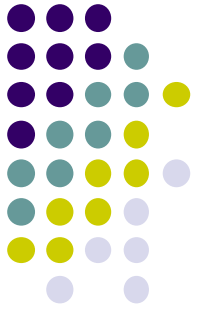
- **Cisco Express Forwarding (CEF)**
  - The preferred Cisco IOS packet-forwarding mechanism. Like fast switching, CEF builds a Forwarding Information Base (FIB), and an adjacency table.
  - The FIB contains pre-computed reverse lookups, next hop information for routes including the interface and Layer 2 information.

# Packet Forwarding



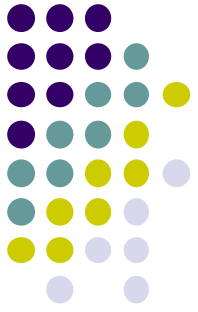
## Cisco Express Forwarding





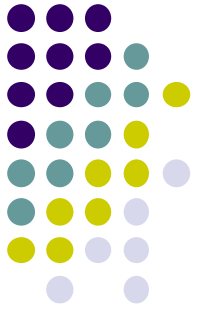
# Packet Forwarding

- Packet Forwarding Decision
  - **Forwards the Packet to a Device on a Directly Connected Network**
  - If the route entry indicates that the egress interface is a directly connected network, this means that the destination IP address of the packet belongs to a device on the directly connected network. Therefore, the packet can be forwarded directly to the destination device.



# Packet Forwarding

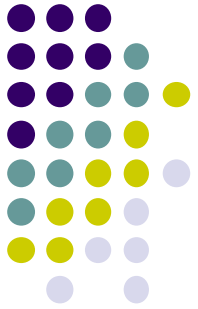
- **Forwards the Packet to a Next-Hop Router**
- If the route entry indicates that the destination IP address is on a remote network, this means the destination IP address of the packet belongs to a device on network that is not directly connected.
- Therefore, the packet must be forwarded to another router, specifically a next-hop router. The next-hop address is indicated in the route entry.



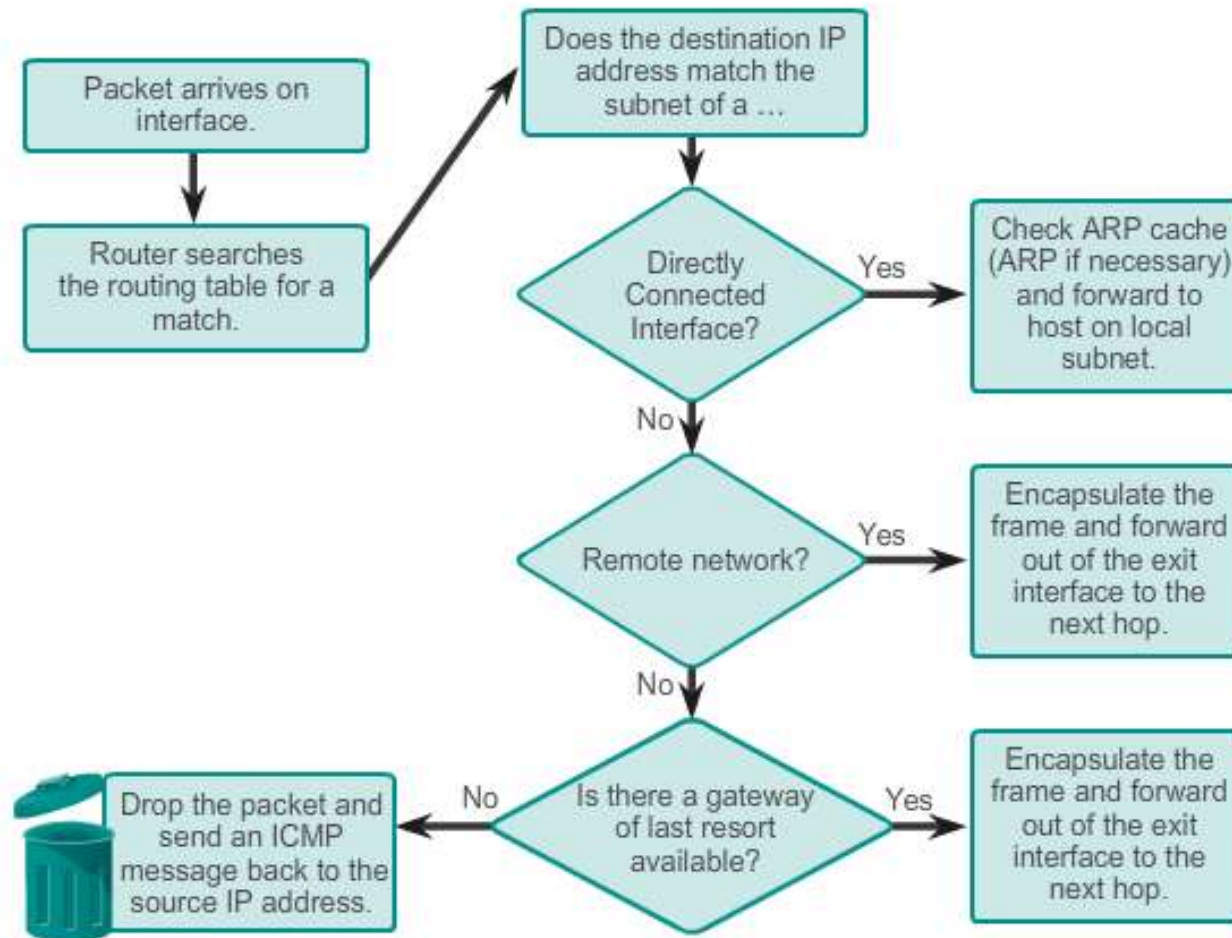
# Packet Forwarding

- **Forwards the Packet to default route**
- If no match, the packet will be forwarded to default route.
- **Drops the Packet - No Match in Routing Table**
- If there is no match between the destination IP address and a prefix in the routing table, and if there is no default route, the packet will be dropped.

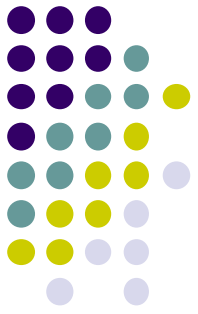
# Packet Forwarding



Packet Forwarding Decision Process



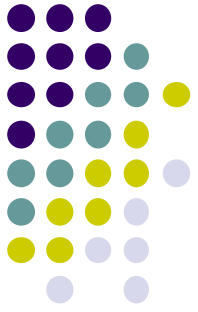




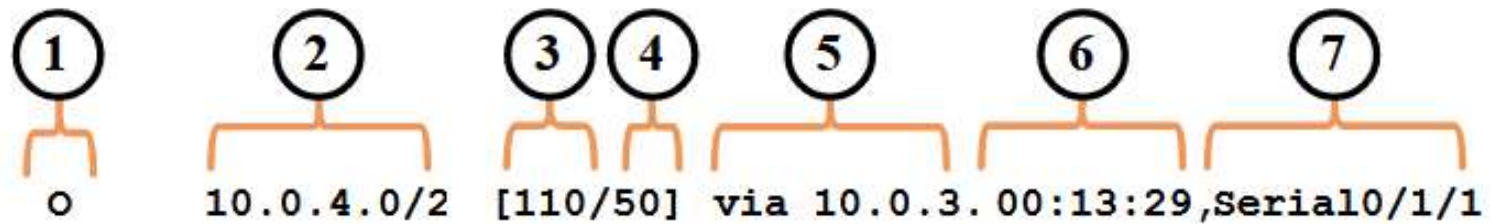
# Routing Table

- Each route entry identifies the following information:
  - **1. Route source** - Identifies how the route was learned.
  - **2. Destination network**
  - **3. Administrative distance**
  - **4. Metric**
  - **5. Next-hop** - Identifies the IP address of the next router to forward the packet to.
  - **6. Route timestamp** - Identifies how much time has passed since the route was learned.
  - **7. Outgoing interface** - Identifies the exit interface to use to forward a packet toward the final destination.

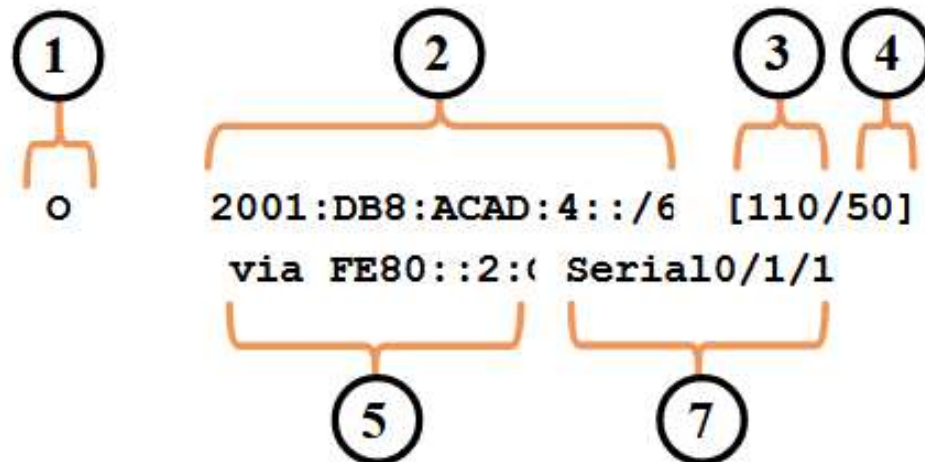
# Routing Table



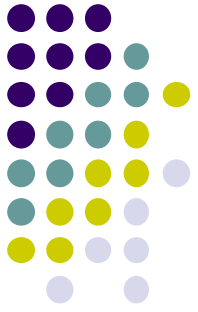
## IPv4 Routing Table



## IPv6 Routing Table

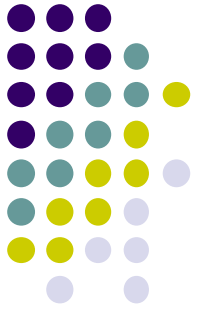


# Routing Table



```
R1# show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
(Output omitted)
C      10.0.1.0/24 is directly connected, GigabitEthernet0/0/0
L      10.0.1.1/32 is directly connected, GigabitEthernet0/0/0
R1#
R1# show ipv6 route
IPv6 Routing Table - default - 10 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
(Output omitted)

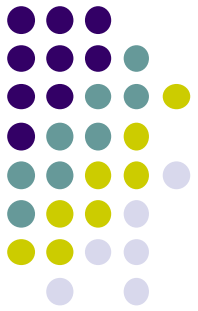
C      2001:DB8:ACAD:1::/64 [0/0]
      via GigabitEthernet0/0/0, directly connected
L      2001:DB8:ACAD:1::1/128 [0/0]
      via GigabitEthernet0/0/0, receive
R1#
```



# Static Route

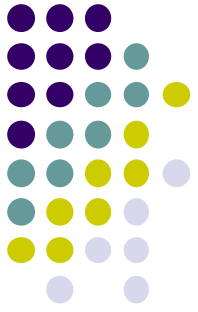
- Static routes are **manually configured**. The routes are not automatically updated and must be manually reconfigured if the network topology changes. A static route is identified in the routing table with the code 'S'.
- The benefits of using static routes include improved security and resource efficiency. Static routes use less bandwidth than dynamic routing protocols, and no CPU cycles are used to calculate and communicate routes.
- The main disadvantage to using static routes is the lack of automatic reconfiguration if the network topology changes.

# Static Route



**Static Routing Advantages and Disadvantages**

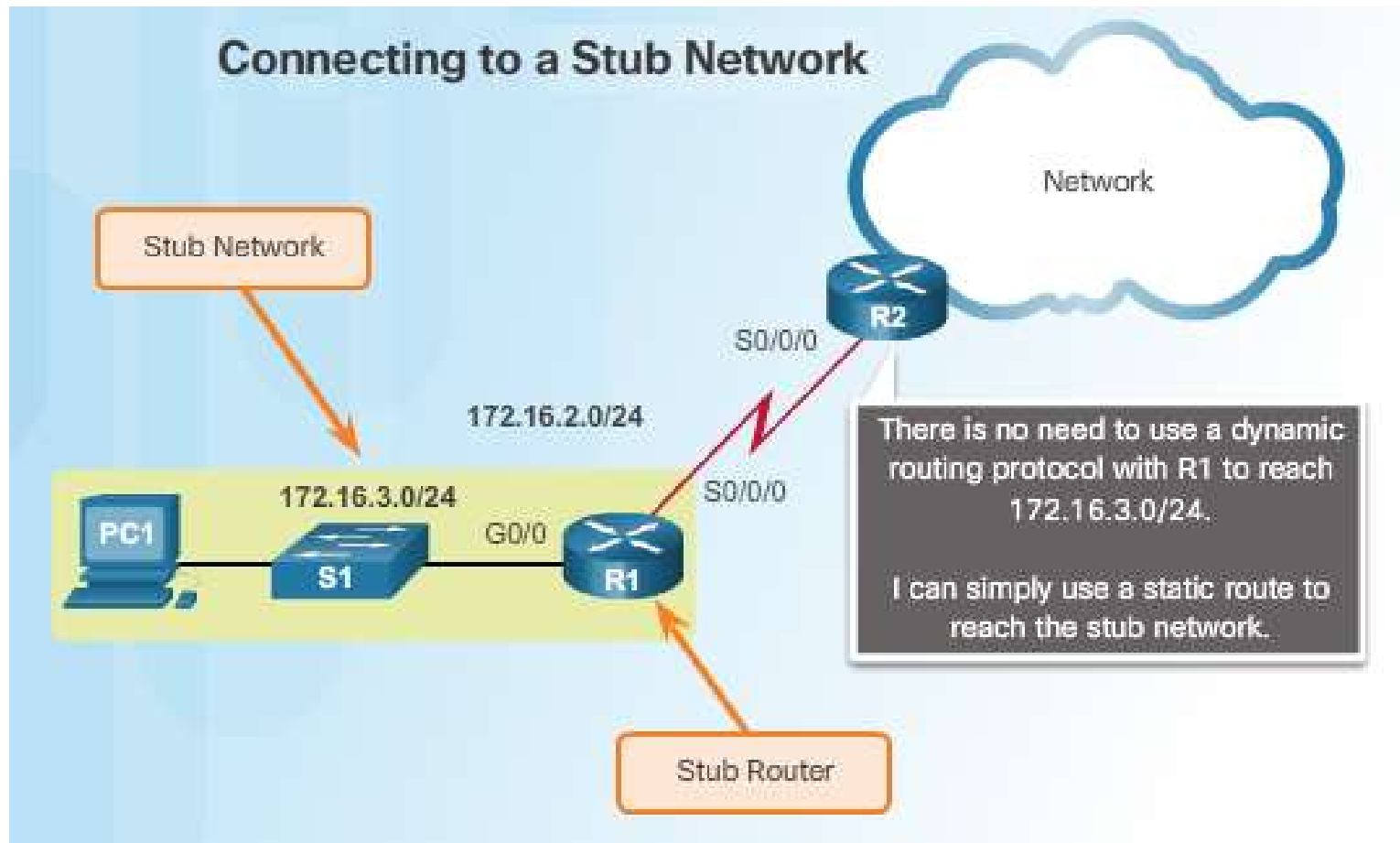
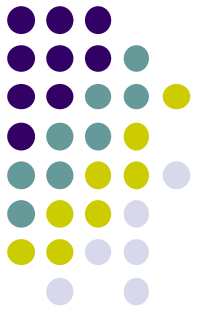
| Advantages                                                                                                   | Disadvantages                                                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Easy to implement in a small network.                                                                        | Suitable only for simple topologies or for special purposes such as a default static route. Configuration complexity increases dramatically as network grows. |
| Very secure. No advertisements are sent as compared to dynamic routing protocols.                            |                                                                                                                                                               |
| Route to destination is always the same.                                                                     | Manual intervention required to re-route traffic.                                                                                                             |
| No routing algorithm or update mechanism required; therefore, extra resources (CPU or RAM) are not required. |                                                                                                                                                               |

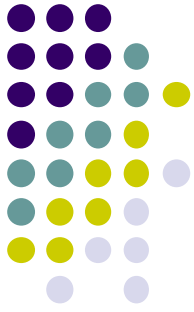


# Static Route

- Static routing has three primary uses:
  - Providing ease of routing table maintenance in smaller networks that are not expected to grow significantly.
  - Routing to and from stub networks. A stub network is a network accessed by a single route, and the router has only one neighbor.
  - Using a single default route to represent a path to any network that does not have a more specific match with another route in the routing table. Default routes are used to send traffic to any destination beyond the next upstream router.

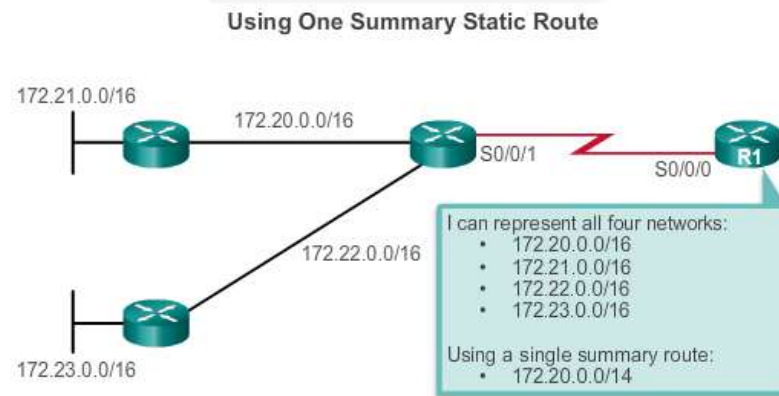
# Static Route





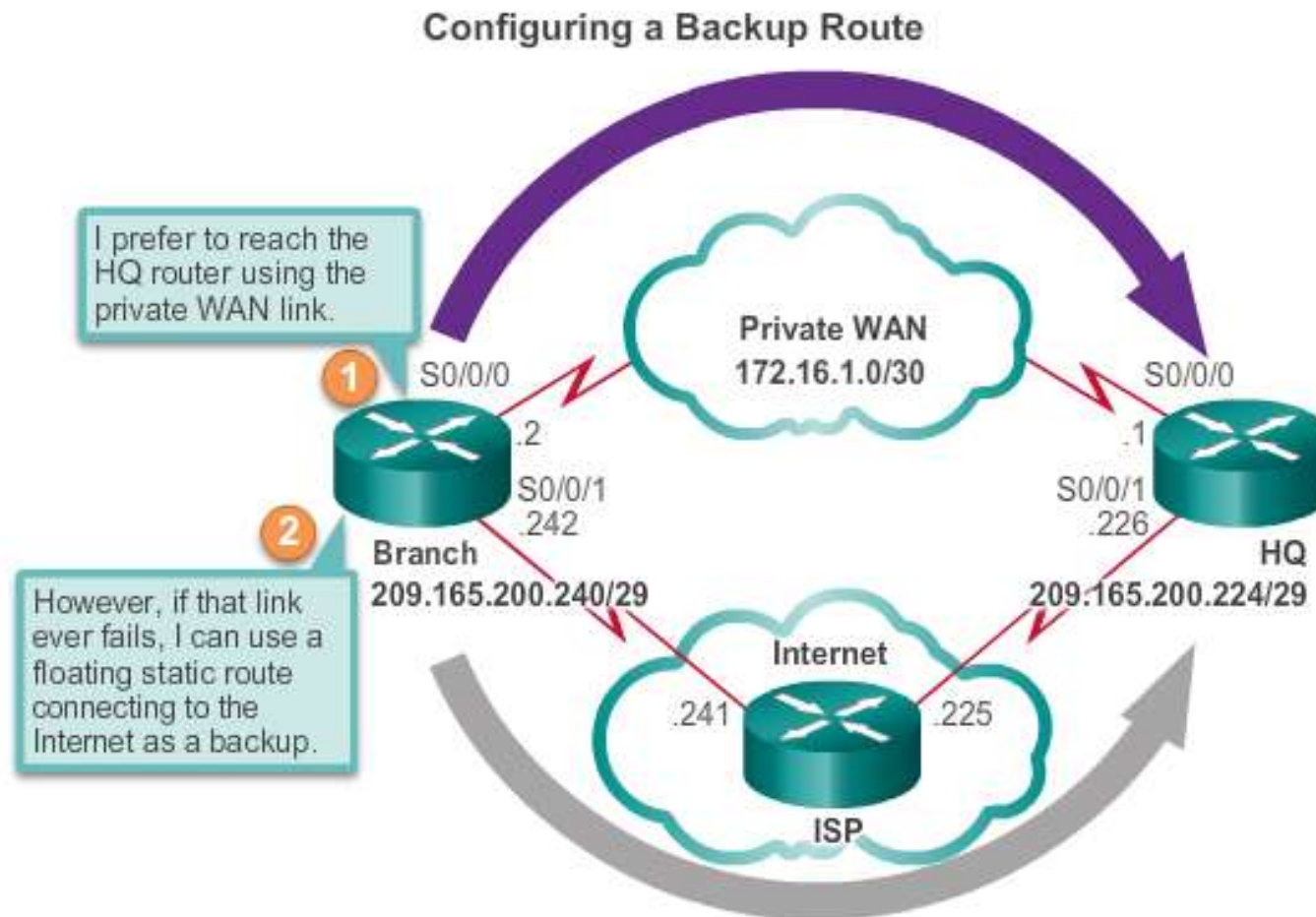
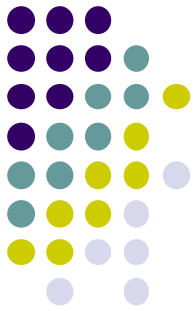
# Static Route

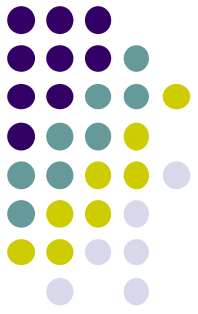
- Static route can also be used to:
  - Reduce the number of routes advertised by summarizing several contiguous networks as one static route
  - Create a backup route in case of a primary route link fails. The backup or floating static route is configured with a higher administrative distance than the primary route.





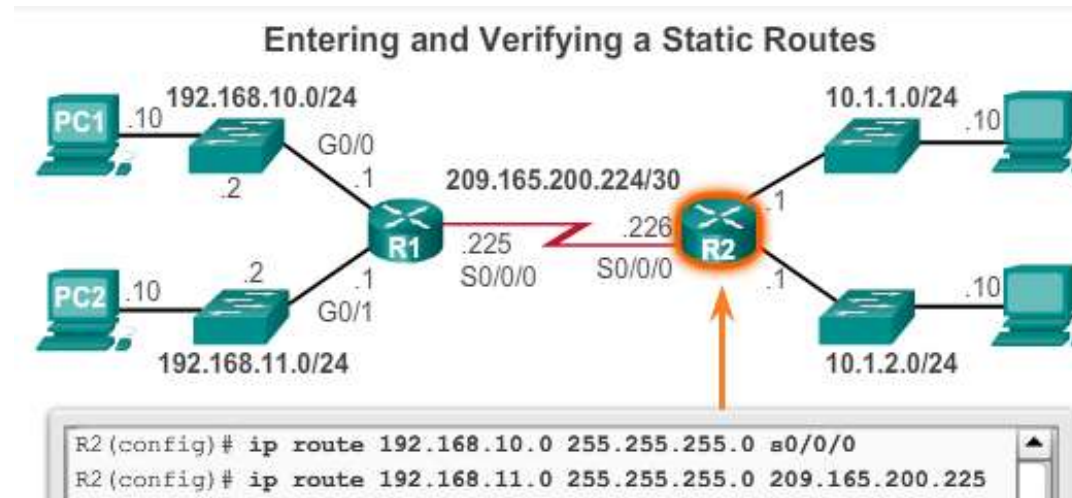
# Static Route



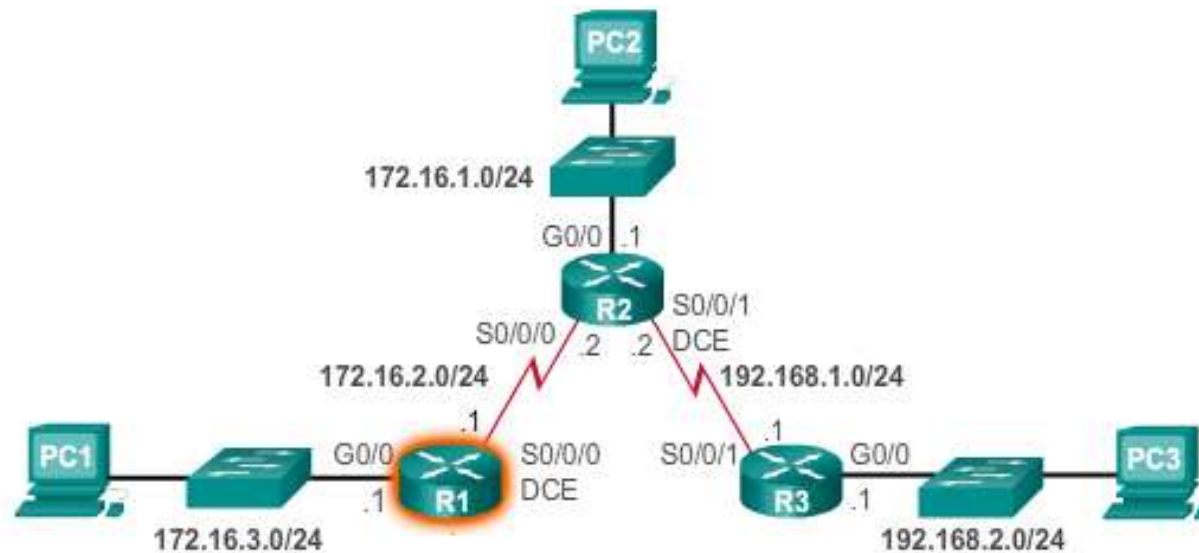
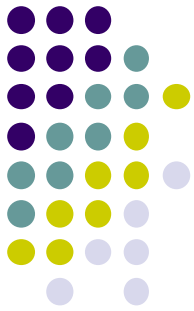


# Static Route

- There are two common types of static routes in the routing table:
  - Static route to a specific network
  - Default static route
- IPv4 static routes are configured using the **ip route** *network mask {next-hop-ip | exit-intf}* global configuration command.



# Static Route



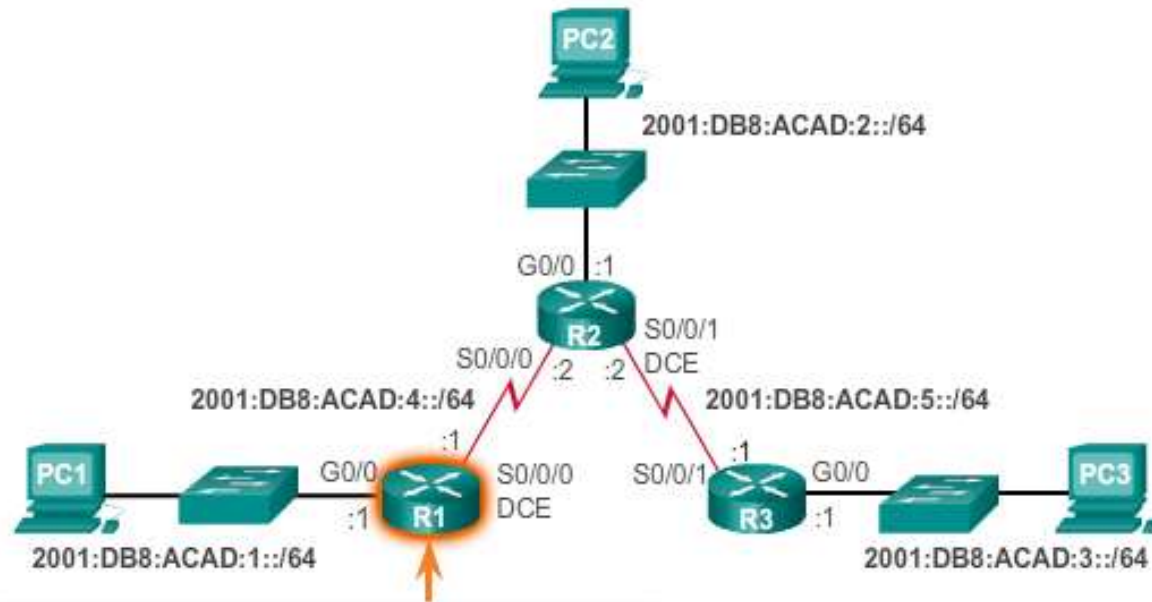
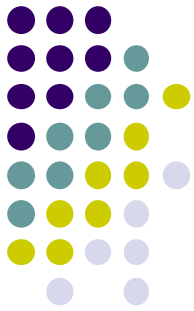
## Configure Directly Connected Static Routes on R1

```
R1(config)# ip route 172.16.1.0 255.255.255.0 s0/0/0
R1(config)# ip route 192.168.1.0 255.255.255.0 s0/0/0
R1(config)# ip route 192.168.2.0 255.255.255.0 s0/0/0
```

## Configuring Next-Hop Static Routes on R1

```
R1(config)# ip route 172.16.1.0 255.255.255.0 172.16.2.2
R1(config)# ip route 192.168.1.0 255.255.255.0 172.16.2.2
R1(config)# ip route 192.168.2.0 255.255.255.0 172.16.2.2
```

# Static Route

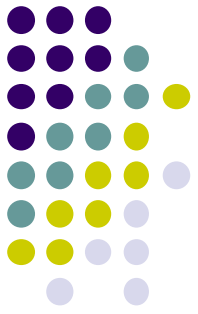


## Configure Next-Hop Static IPv6 Routes

```
R1(config)# ipv6 route 2001:DB8:ACAD:2::/64 2001:DB8:ACAD:4::2
R1(config)# ipv6 route 2001:DB8:ACAD:5::/64 2001:DB8:ACAD:4::2
R1(config)# ipv6 route 2001:DB8:ACAD:3::/64 2001:DB8:ACAD:4::2
```

## Configure Directly Connected Static IPv6 Routes on R1

```
R1(config)# ipv6 route 2001:DB8:ACAD:2::/64 s0/0/0
R1(config)# ipv6 route 2001:DB8:ACAD:5::/64 s0/0/0
R1(config)# ipv6 route 2001:DB8:ACAD:3::/64 s0/0/0
```

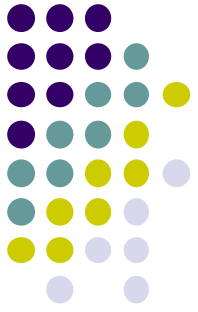


# Static Route

- If the IPv6 static route uses an IPv6 link-local address as the next-hop address, a fully specified static route including the exit interface must be used.
- The reason a fully specified static route must be used is because IPv6 link-local addresses are not contained in the IPv6 routing table. Link-local addresses are only unique on a given link or network.

Configure Fully Specified Static IPv6 Routes on R1



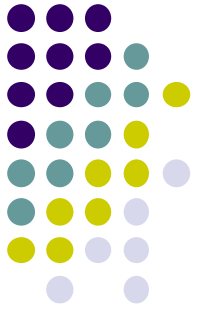


# Static Route

- A fully specified static route can be configured when the next-hop IP address and exit interface are specified using the **ip route** *network mask exit-intf next-hop-ip* command.

```
R1(config)# ip route 172.16.1.0 255.255.255.0 GigabitEthernet 0/0/1 172.16.2.2
R1(config)# ip route 192.168.1.0 255.255.255.0 GigabitEthernet 0/0/1 172.16.2.2
R1(config)# ip route 192.168.2.0 255.255.255.0 GigabitEthernet 0/0/1 172.16.2.2
```

```
R1(config)# ipv6 route 2001:db8:acad:1::/64 s0/1/0 fe80::2
```



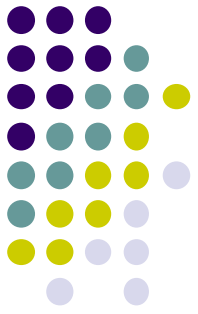
# Static Route

- The default static route specifies the exit point to use when the routing table does not contain a path for the destination network.
- A default static route is useful when a router has only one exit point to another router, such as when the router connects to a central router or service provider.
- Use the **ip route 0.0.0.0 0.0.0.0 {exit-intf | next-hop-ip}** global configuration command.

Configure a default static route using 209.165.200.226 as the next-hop address and exit configuration mode.

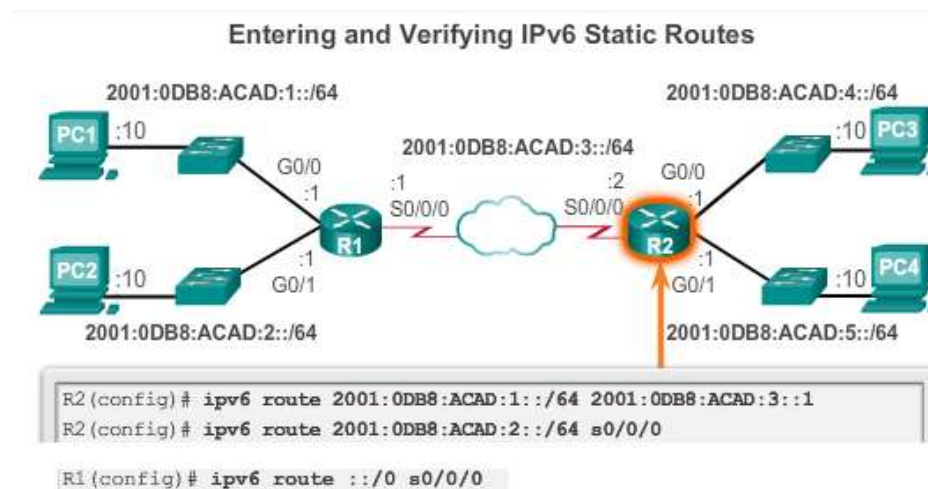
```
R1(config)# ip route 0.0.0.0 0.0.0.0 209.165.200.226
```





# Static Route

- IPv6 supports static and default static routes. Static IPv6 routes are configured using the **ipv6 route** *ipv6-prefix/prefix-length {ipv6-address|interface-type interface-number}* global configuration command.
- To configure a default static IPv6 route, use the **ipv6 route ::/0** *{ipv6-address | interface-type interface-number}* global configuration command.

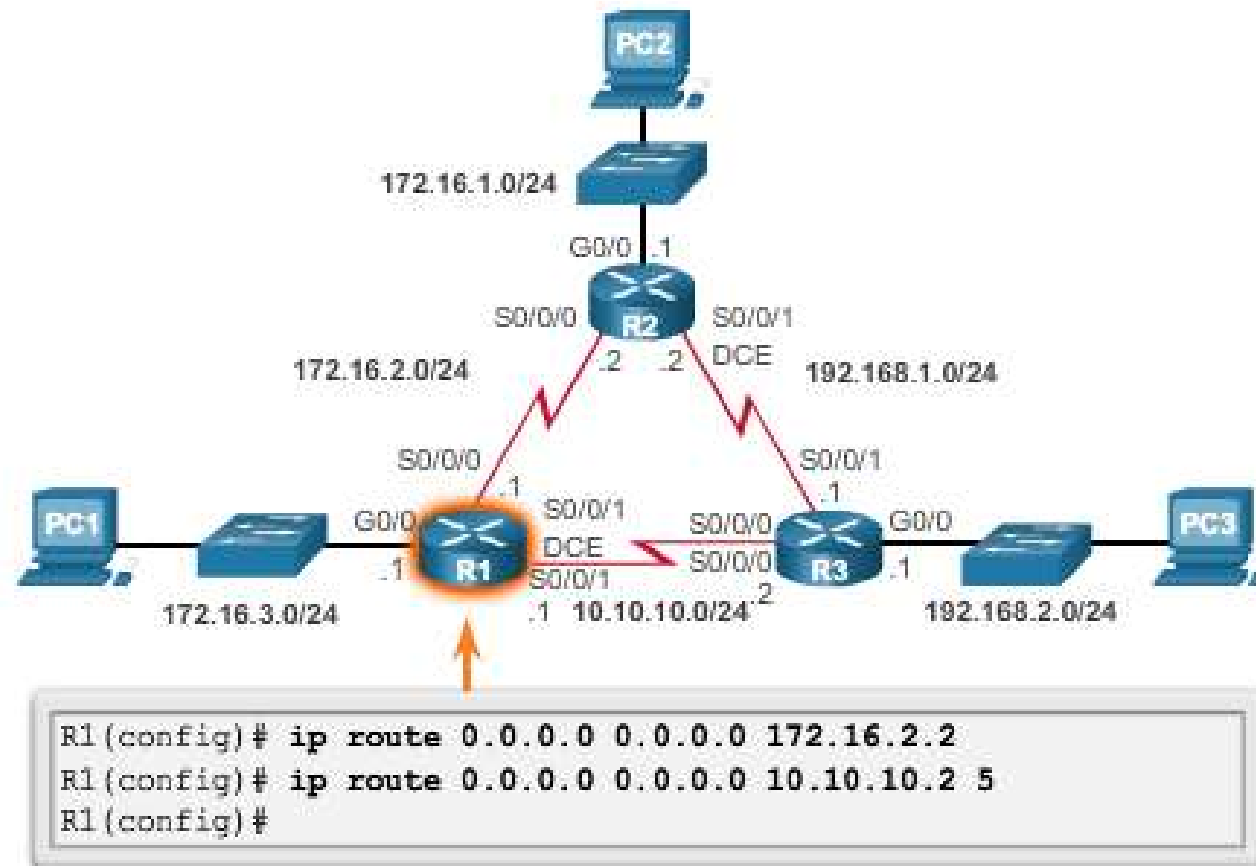


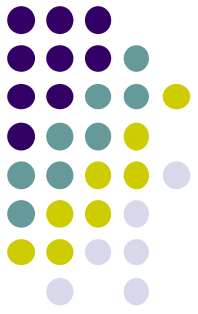


# Static Route

- Configuring floating static route

Configuring a Floating Static Route to R3

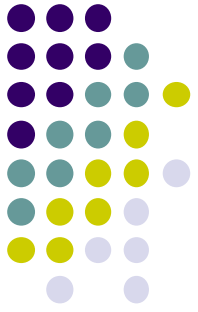




# Static Route

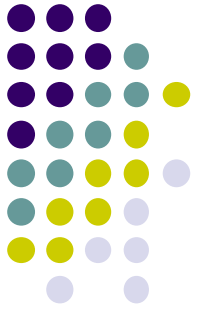
- A host route is an IPv4 address with a 32-bit mask, or an IPv6 address with a 128-bit mask. A host route allows for a more efficient process for packets that are directed to the router itself, rather than for packet forwarding.
- When an active interface on a router is configured with an IP address, a local host route is automatically added to the routing table. The local routes are marked with **L** in the output of the routing table.

```
Branch(config)# ip route 209.165.200.238 255.255.255.255 198.51.100.2
Branch(config)# ipv6 route 2001:db8:acad:2::238/128 2001:db8:acad:1::2
Branch(config)# exit
Branch#
```



# Static Route

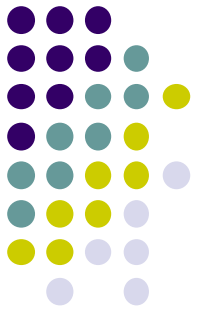
- Common IOS troubleshooting commands include the following:
  - ping
  - traceroute
  - show ip route
  - show ip interface brief
  - show cdp neighbors detail



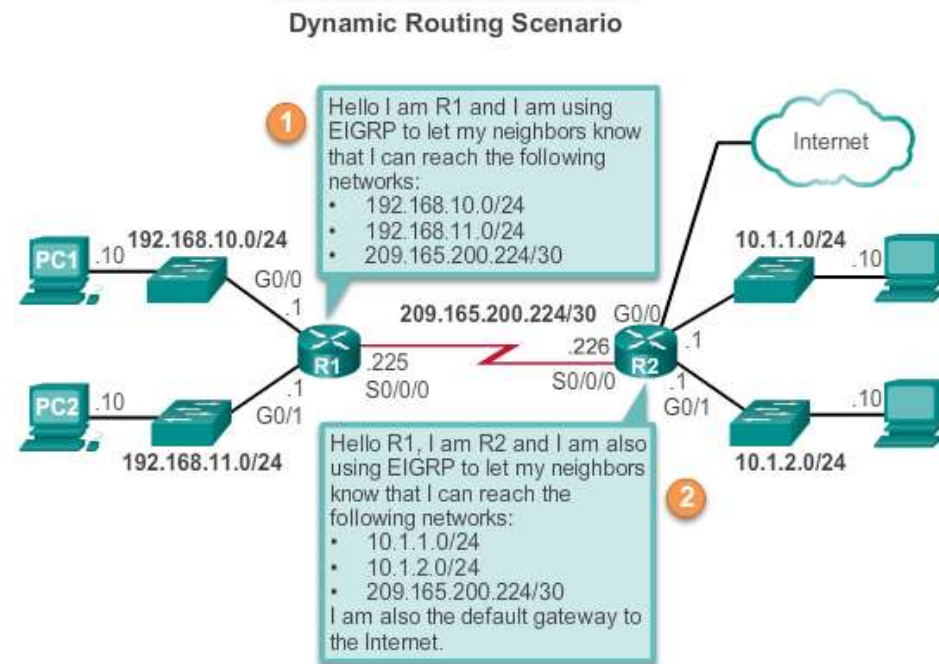
# Dynamic Route

- Dynamic routing protocols are used by routers to automatically share information about the reachability and status of remote networks.
- Dynamic routing protocols perform several activities, including network discovery and maintaining routing tables.
- The advantages of dynamic routing protocols are the ability to select a best path, and the ability to automatically discover a new best path when there is a change in the topology.

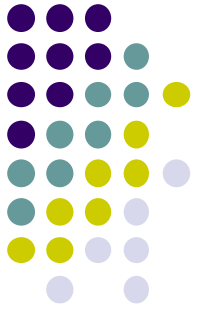
# Dynamic Route



- During network discovery, routers exchange routes and update their routing tables. Routers have converged after they have finished exchanging and updating their routing tables.

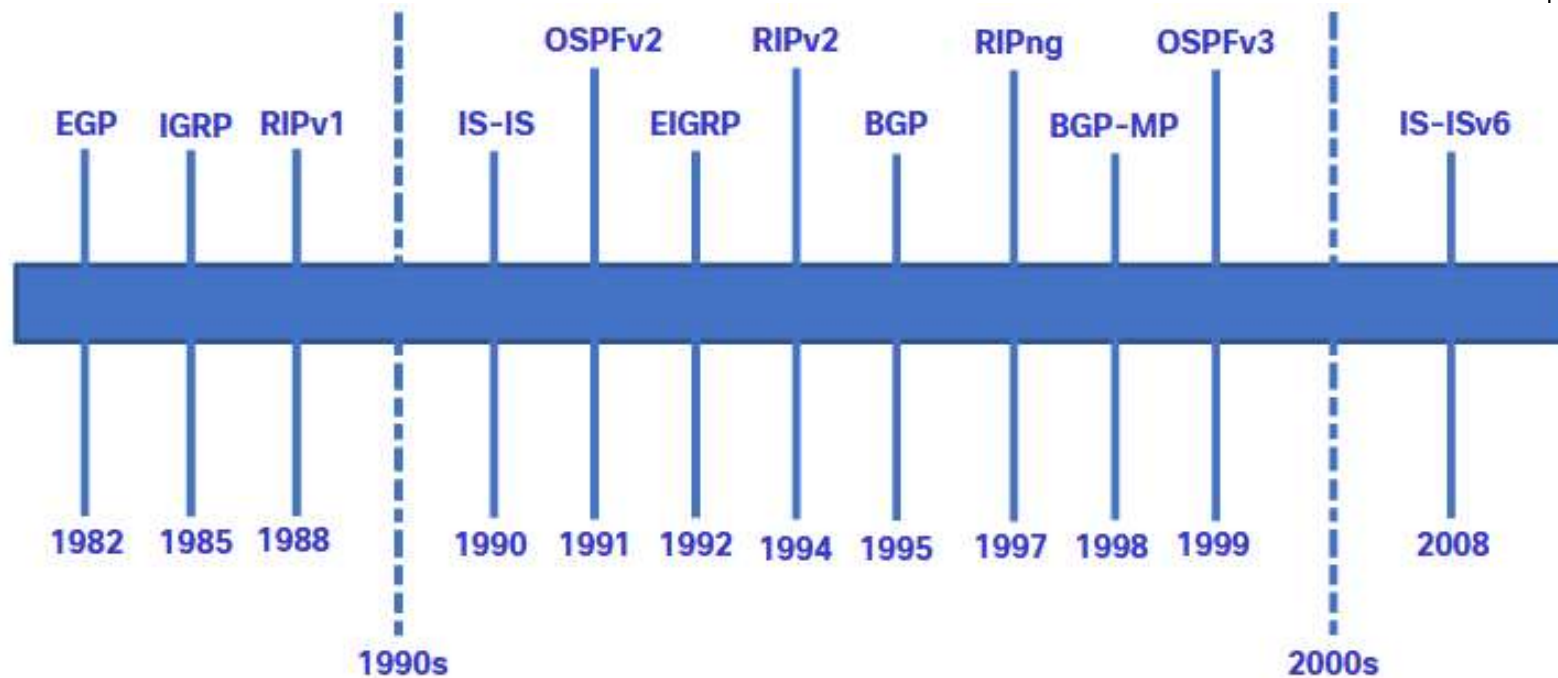
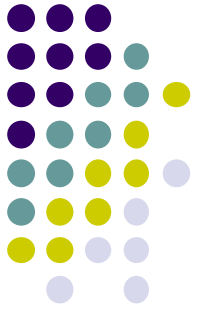


# Dynamic Route

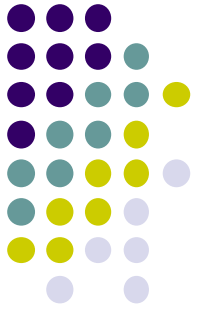


| Feature                  | Dynamic Routing                                     | Static Routing                          |
|--------------------------|-----------------------------------------------------|-----------------------------------------|
| Configuration complexity | Independent of network size                         | Increases with network size             |
| Topology changes         | Automatically adapts to topology changes            | Administrator intervention required     |
| Scalability              | Suitable for simple to complex network topologies   | Suitable for simple topologies          |
| Security                 | Security must be configured                         | Security is inherent                    |
| Resource Usage           | Uses CPU, memory, and link bandwidth                | No additional resources needed          |
| Path Predictability      | Route depends on topology and routing protocol used | Explicitly defined by the administrator |

# Dynamic Route



|      | Interior Gateway Protocols |                |            |                | Exterior Gateway Protocols |
|------|----------------------------|----------------|------------|----------------|----------------------------|
|      | Distance Vector            |                | Link-State |                | Path Vector                |
| IPv4 | RIPv2                      | EIGRP          | OSPFv2     | IS-IS          | BGP-4                      |
| IPv6 | RIPng                      | EIGRP for IPv6 | OSPFv3     | IS-IS for IPv6 | BGP-MP                     |



# Dynamic Route

- The main components of dynamic routing protocols include:
- **Data structures** - Routing protocols typically use tables or databases for its operations. This information is kept in RAM.
- **Routing protocol messages** - Routing protocols use various types of messages to discover neighboring routers, exchange routing information, and other tasks to learn and maintain accurate information about the network.
- **Algorithm** - Routing protocols use algorithms for facilitating routing information and for best path determination.